

minimise the device parasitics for high-performance operation. On this layer, a suitable catalyst is deposited to enhance growth and directionality. Then the carbon nanotubes are grown in a CVD furnace. Source and drain fingers and pads are then defined by the use of standard photolithography. Then a gate oxide is grown, on top of which the gate is patterned.

Source-to-drain distance, or channel width, is $0.8\text{ }\mu\text{m}$. Gate length is about $0.45\text{ }\mu\text{m}$ and total gate periphery is $800\text{ }\mu\text{m}$. The amplifier based on CNFET is typically biased at $V_{GS} = -0.5\text{V}$ and $V_{DS} = 2.5\text{V}$. Gate and drain DC bias are applied through choke. About 12.5 dB of gain can be obtained from a single stage centered at 1.3 GHz as shown in Fig. 7.

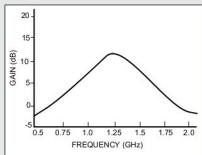


Fig. 7: CNFET amplifier gain

5 Carbon-nanotube pressure sensor

There has been much interest over the last couple of decades in exploiting the outstanding electrical, mechanical and optical properties of carbon nanotubes to fabricate micro-electromechanical system (MEMS) devices. In 2010, a carbon nanotube forest-based pressure sensor was demonstrated that had a nearly symmetrical response for both positive and negative gauge pressures, using a suspended diaphragm entirely covered with CNT forest.

Achieving reliable devices based on individual CNTs with high

repeatability is still very challenging, because of the structural variation between individual nanotubes. This problem can be addressed by using CNT forests—a macroscopic network of virtually self-aligned CNTs which have anisotropic electromechanical properties that can be used in many MEMS applications.

CNT forests provide a very large surface area and exhibit piezoresistivity when a lateral stress is applied. This property has been utilised to create a very sensitive and accurate physical sensor for pressure measurement. The sensor is fabricated by a deflectable, $8\text{ }\mu\text{m}$ thick parylene—a biocompatible polymer membrane—and hence is of particular interest for medical applications.

The membrane is suspended by a silicon frame. The sensitivities to positive and negative gauge pressures are found to be comparable in magnitude with the average values of -986 and $+816\text{ ppm/kPa}$, respectively.

As shown in Fig. 8, multiwalled CNT forest is supported on a $5\times 5\text{ mm}^2$ membrane of parylene-C of $8\text{ }\mu\text{m}$ thickness. The two opposing ends of the forest are connected with the 130 nm thick Mo metal pads on the substrate that are used to measure the

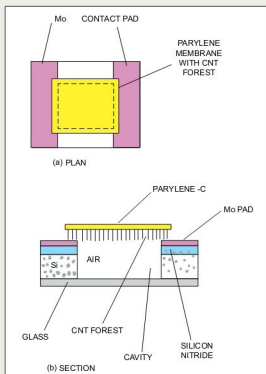


Fig. 8: CNT pressure sensor

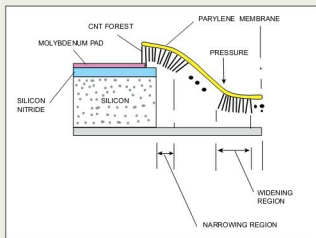


Fig. 9: Operation of CNT pressure sensor

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